Third étale homotopy groups are abelian

Disproof of Conjecture 00000002440

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Abstract

Every third homotopy group is abelian by the Eckmann–Hilton argument. Applying this to each pointed space in an étale homotopy type shows that its third homotopy pro-group has an abelian-valued representative. Thus no smooth variety can have the internally nonabelian third étale homotopy group requested in the conjecture. Lean verifies the obstruction for actual cube-loop homotopy groups and entire diagrams of those groups. The standard étale-type and realization constructions connecting this universal result to varieties are explicitly stated as mathematical definition bridges outside the Lean formalization.

The obstruction for actual pointed spaces

Let (Y, y) be any pointed topological space. Represent $\pi_3(Y, y)$ by continuous maps

$$f : [0,1]^3 \longrightarrow Y, \qquad f(\partial[0,1]^3) = \{y\},$$

modulo homotopy relative to the entire boundary. Concatenating in either of the first two coordinate directions gives two operations m_0, m_1 on these homotopy classes. Both have the constant class e as a two-sided unit, and their actual coordinate concatenations satisfy interchange:

$$m_0\big(m_1(a,b), m_1(c,d)\big) = m_1\big(m_0(a,c), m_0(b,d)\big).$$

The interchange identity holds for the concrete cube maps; the unit identities hold on relative-boundary homotopy classes. Substitution gives

$$\begin{aligned} m_0(a,b) &= m_0\big(m_1(a,e), m_1(e,b)\big) \\ &= m_1\big(m_0(a,e), m_0(e,b)\big) = m_1(a,b), \\ m_0(a,b) &= m_0\big(m_1(e,a), m_1(b,e)\big) \\ &= m_1\big(m_0(e,b), m_0(a,e)\big) = m_1(b,a). \end{aligned}$$

The operations coincide and commute. In particular, the genuine group $\pi_3(Y, y)$ is abelian for every pointed space.

Application to the étale homotopy pro-group

For a pointed locally noetherian scheme $(X, \bar{x})$, the standard Artin–Mazur–Friedlander étale homotopy type is a pointed pro-object in the homotopy category of simplicial sets. Its higher étale homotopy pro-groups are obtained by applying the ordinary pointed homotopy-group functor; see [2], §2.1. Choose a pointed simplicial representative at each level and use geometric realization. Its third group is the actual group

$$\pi_3\big(|K_i|, k_i\big).$$

Each of these groups is abelian by the previous argument. Every transition homomorphism is consequently a homomorphism of abelian groups. Hence the entire defining diagram factors through abelian groups before any inverse limit is taken. The étale third homotopy pro-group is therefore a pro-abelian group.

Theorem 1. *Under the standard étale homotopy-group definition, no smooth variety X has internally nonabelian $\pi_3^{\text{ét}}(X, \bar{x})$. Consequently the variety required by Conjecture 00000002440 does not exist.*

Proof. A smooth variety is locally noetherian. Its étale homotopy type and third homotopy pro-group have the representation just described, with all groups abelian. This is incompatible with internal noncommutativity. The additional condition that $\pi_1^{\text{ét}}$ be finite and the proposed Enriques-type construction cannot change this universal obstruction. □

Pro-objects and profinite conventions. The conclusion is that the pro-group has an abelian-valued representative. It does not assert that every arbitrary pro-isomorphic presentation has abelian groups at every level. Also, an abelian inverse limit alone would not establish the required statement about a pro-group; the argument above is levelwise and gives the entire diagram factorization.

If profinite completion is included in the convention, it preserves commutativity. Indeed every quotient image of an abelian group is abelian. A compatible-tuple inverse limit of abelian groups is a subgroup of their product with pointwise multiplication, which is again commutative. These observations are additional to, rather than replacements for, the levelwise pro-group argument.

A possibly nontrivial action of π_1 on π_3 is a separate structure. Such an action cannot make the internal group operation on π_3 noncommutative, as explicitly required by both versions of the conjecture.

What the Lean project verifies

The input `BasedSpace` contains only a carrier, a topology and a base point. Its `Pi3` is Mathlib's actual `HomotopyGroup.Pi 3`, the relative-boundary cube-loop homotopy quotient. No group or commutativity hypothesis is part of the input. Mathlib's proved Eckmann–Hilton construction supplies its commutative group structure. The submitted file exposes the actual distinct-coordinate interchange identity and the two unit laws; it also proves that reversing concatenation gives actual cube maps homotopic relative to the boundary.

A diagram has actual pointed spaces at its objects. Its arrows are homomorphisms of their actual third homotopy groups, with identity and composition laws. It assumes no commutativity. For every indexing category the project constructs an explicit natural isomorphism between its group-valued diagram and the forgetful image of a diagram in `CommGrp`. Cofiltered diagrams are included. Thus the checked result handles every transition diagram that can arise in the stated application. It does not infer pro-commutativity from an inverse-limit calculation.

Further theorems prove commutativity of arbitrary surjective quotient images and normal quotients of actual third homotopy groups. They construct the subgroup of all transition-compatible tuples and prove its multiplication commutative, for original groups and quotient diagrams alike.

Explicit formal boundary. Lean does not construct schemes, étale hypercovers, the étale type functor, geometric realization, its homotopy-group comparison, or the homomorphism functor induced by pointed maps. Those standard constructions are the mathematical bridges on page 2. The universal theorem for every actual pointed space and every homomorphism diagram is proved in Lean. No custom object is merely named an étale group and then assumed commutative.

Reproduction. Lean 4.19.0 is pinned. The Mathlib revision is

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

In `lean/`, run:

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

Compile with `tectonic main.tex`. Actual fresh-build logs, hashes and eight printed axiom reports are included; only standard foundational axioms occur. No numerical oracle is used. The work received author, parent and internal cross-review; no external independent review is claimed.

References

[1] The Last Math Competition, Conjecture 00000002440. Exact bilingual source included.

[2] M. Moutand, *On schemes with trivial higher étale homotopy groups*, arXiv:2312.08505v2 (26 February 2026), §2.1.